Dongki Kim

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Research Interest

My research centers on the intersection of AI and the life sciences. I aim to develop AI systems that can understand and reason about complex biological systems, including representation learning, generative models, LLMs, and agents, with the goal of accelerating scientific discovery.

Education

KAIST — **Daejeon, South Korea**
Ph.D. in Artificial Intelligence — *Sep. 2023 – Present*
M.S. in Artificial Intelligence — *Sep. 2021 – Aug. 2023*
- Advisor: Prof. Sung Ju Hwang

Seoul National University (SNU) — **Seoul, South Korea**
B.S. in Computer Science and Engineering — *Mar. 2014 – Feb. 2021*
B.S. in Applied Biology and Chemistry — *Mar. 2014 – Feb. 2021*

Publications

Principia: Test-Time Hypotheses to Principle Construction for Scientific Discovery
Dongki Kim*, Sangwoo Park*, Jinheon Baek, Soyeong Jeong, Sung Ju Hwang
arXiv preprint, 2026

AutoSciBench: Autonomous Benchmark Generation for Evaluating Scientific Agents
Dongki Kim, Namkyeong Lee, Surag Nair, Carl Edwards, Xiner Li, Edward De Brouwer, Jenna Lynn Collier, Sung Ju Hwang, Gabriele Scalia, Ehsan Hajiramezanali
arXiv preprint, 2026

AdaPath: Query-Adaptive Path-Finding via Path-Bank for Multi-Hop Implicit Biomedical KGQA
Jun Hyeong Kim, Dongki Kim, Yinhua Piao, Sung Ju Hwang
Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2026

ES-Merging: Biological MLLM Merging via Embedding Space Signals
Wonbin Lee*, Dongki Kim*, Sung Ju Hwang
Conference on Neural Information Processing Systems (**NeurIPS**), 2026

Rethinking Reward Models for Multi-Domain Test-Time Scaling
Dong Bok Lee*, Seanie Lee*, Sangwoo Park, Minki Kang, Jinheon Baek, Dongki Kim, Dominik Wagner, Jiongdao Jin, Heejun Lee, Tobias Bocklet, Jinyu Wang, Jingjing Fu, Sung Ju Hwang, Jiang Bian, Lei Song
Transactions on Machine Learning Research (**TMLR**), 2026
Conference on Neural Information Processing Systems (**NeurIPS**), 2026

Multimodal Prompt Optimization: Why Not Leverage Multiple Modalities for MLLMs
Yumin Choi*, Dongki Kim*, Jinheon Baek, Sung Ju Hwang
International Conference on Learning Representations (**ICLR**), 2026

Mol-LLaMA: Towards General Understanding of Molecules in Large Molecular Language Model
Dongki Kim, Wonbin Lee, Sung Ju Hwang
Conference on Neural Information Processing Systems (**NeurIPS**), 2025
AI for Science Workshop at NeurIPS (**AI4Science @ NeurIPS**), 2025

Graph Generation with Diffusion Mixture
Jaehyeong Jo*, Dongki Kim*, Sung Ju Hwang
International Conference on Machine Learning (**ICML**), 2024
ML for Drug Discovery Workshop at ICLR (**MLDD @ ICLR**), 2023 **(Spotlight)**

Protein Representation Learning by Capturing Protein Sequence-Structure-Function Relationship
Eunji Ko*, Seul Lee*, Minseon Kim*, Dongki Kim, Sung Ju Hwang
ML for Genomics Explorations Workshop at ICLR (**MLGenX @ ICLR**), 2024 **(Spotlight)**

Antibody-SGM: Antigen-Specific Joint Design of Antibody Sequence and Structure using Diffusion Models
Xuezhi Xie, Jin Sub Lee, Dongki Kim, Jaehyeong Jo, Jisun Kim, Philip M. Kim
Computational Biology Workshop at ICML (**CompBio @ ICML**), 2023

Graph Self-supervised Learning with Accurate Discrepancy Learning
Dongki Kim*, Jinheon Baek*, Sung Ju Hwang
Conference on Neural Information Processing Systems (**NeurIPS**), 2022

Edge Representation Learning with Hypergraphs
Jaehyeong Jo*, Jinheon Baek*, Seul Lee*, Dongki Kim, Minki Kang, Sung Ju Hwang
Conference on Neural Information Processing Systems (**NeurIPS**), 2021
(∗: equal contribution)

Research Experience

Research Intern, Genentech *Jun. 2026 – Aug. 2026*
- Conducting research on automatic pipeline construction for evaluating state-of-the-art agents.

Graduate Research Assistant, KAIST (Advisor: Prof. Sung Ju Hwang) *Mar. 2021 – Present*
- Conducting research on representation learning, generative model, LLMs and agent for graph-structured data with the application to the molecular and general graphs.

Visiting Student, University of Toronto (Host: Prof. Philip M. Kim) *Feb. 2023 – Feb. 2023*
- Conducted research on antigen-specific antibody design via diffusion models.

Talks and Seminars

Towards General Understanding of Molecules in Large Molecular Language Model
at KAIST *May. 2025*

Explainable PK/Tox Prediction: Molecular-Protein-Language Tri-modality Foundation Model
at Korea Machine Learning Ledger Orchestration for Drug Discovery *March. 2025*

Generation of Graph-Structured Data with Diffusion Models
at University of Toronto *Feb. 2023*

Graph Self-supervised Learning with Accurate Discrepancy Learning
at KAIST *Nov. 2022*

Academic Services

Reviewer
- *NeurIPS* (2022–2026)
- *ICML* (2022–2026)
- *ICLR* (2024–2027)
- *AAAI* (2027)
- *TMLR* (2025)
- *GEM @ ICLR* (2024–2025)

Reference

- Prof. Sung Ju Hwang, Endowed Chair Professor, KAIST
  E-mail: sungju.hwang@kaist.ac.kr